

Basic Machine Learning I

Objectives, datasets, losses, and baselines

Jonathon Hare

What makes a problem supervised?

Inputs, targets, models, and predictions

From individual errors to empirical risk

Regression losses encode different assumptions

Classification requires scores and probabilities

Binary and multiclass cross-entropy

Baselines reveal whether learning is useful

Training, validation, and test data have different jobs

Dataset construction shapes the result

A complete supervised-learning workflow